SPE-229088-MS

From Flare to Power: Field-Proven Mobile LNG Technology for Decarbonizing Oilfield and Off-Grid Energy Use Cases

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

This paper introduces F2X, a first-of-its-kind, field-deployable, trailer-mounted liquefaction system engineered to convert flared or stranded gas into low-carbon LNG. The objective is to evaluate how this mobile LNG platform can mitigate methane emissions while displacing diesel across oilfield service operations—including drilling and hydraulic fracturing—as well as in off-grid EV charging and heavy-duty transportation. The scope includes lifecycle emissions benchmarking and the system's scalability across diverse geographies and use cases.

Field trials conducted in Texas confirmed continuous LNG production with zero flaring or venting. A third-party lifecycle analysis using the GREET (2023) model and aligned with ISO 14067:2018 verified the LNG's carbon intensity at -43 kg CO_2e/MMbtu on a well-to-pump basis, with up to 20,000 tCO_2e/year offset per unit when displacing diesel. Compared to centralized LNG solutions, F2X offers an order-of-magnitude reduction in capital cost and deployment time without requiring subsidies or gas grid connectivity.

This paper reframes flare gas utilization as a new class of distributed, low-carbon infrastructure—scalable, modular, and compatible with emerging clean fuel regulations. It delivers a practical blueprint for upstream and distributed energy decarbonization.

Introduction

Routine flaring of associated gas remains a persistent issue in upstream oil and gas operations, particularly in remote or infrastructure-limited regions. Over 140 billion cubic meters of gas are flared globally each year (World Bank, 2024), contributing significantly to methane emissions and lost energy value. This challenge is especially pronounced at stranded wells—locations with no gas gathering pipeline—or in regions where central gas treatment is infeasible.

The global regulatory landscape is rapidly evolving. Commitments under OGMP 2.0, enhanced methane fees in the U.S. Inflation Reduction Act, and tightening EU import standards are all accelerating the need for real, field-deployable solutions. Oil producers now face a dual challenge: sustaining production while meeting aggressive decarbonization and emissions compliance goals.

The environmental and economic case for flare gas recovery has been studied extensively, highlighting both the global mitigation potential and deployment challenges at scale (El-Halwagi et al., 2020). In these scenarios, even when flare gas could theoretically be captured and used, three core challenges hinder investment:

- Volume Variability: Associated gas production often fluctuates with oil flow, leading to unpredictable availability.
- Uncertain Duration: Wells may have short production horizons, limiting return on fixed infrastructure.
- High CAPEX Thresholds: Centralized solutions require long asset life and steady volumes to justify cost.

Infield power generation—using conditioned gas to feed gas engines or turbines—is often a partial solution. However, this approach depends on being able to consume all generated power on-site, which is rarely practical. Many pads do not have enough power demand to match the gas available, and the lack of grid export options leaves operators with limited alternatives.

The F2X platform was developed to solve this exact gap. By converting flare gas into LNG directly at the wellsite, F2X transforms excess gas into a dense, transportable, and storable energy product. This enables:

- Off-site use in applications such as industrial power and transportation
- Diesel displacement within a regional radius (e.g., drilling rigs, remote operations)
- Emissions reduction and compliance with flaring regulations

Unlike infield gas engines that require immediate and total consumption, LNG allows flexible timing and distribution of energy use. This decouples gas capture from power demand, unlocking broader value for stranded gas resources while reducing environmental impact. For oil producers, F2X creates a monetization pathway where none previously existed.

Understanding Flare Gas

Flare gas broadly refers to any gas that is burned off as waste, typically due to the absence of infrastructure to process, treat, or transport it. In the context of oil wells, flaring only occurs when oil is being produced but the associated gas cannot be monetized—often because there is no pipeline or facility available to handle it. Therefore, the majority of flare gas relevant to this paper is associated gas.

Associated gas is the portion of natural gas that is co-produced alongside crude oil. It may be dissolved in the oil under reservoir pressure ("solution gas") or exist as a free gas cap above the oil column. In both forms, it is brought to the surface during oil production. Unlike non-associated or "dry" gas, associated gas tends to have a highly variable flow rate that depends on oil production conditions—such as choke settings, reservoir pressure decline, artificial lift operations, and water cut. Moreover, it is typically much "richer" in heavier hydrocarbons like ethane, propane, butanes, and pentanes. These components increase the gas's energy content but also raise the hydrocarbon dew point and decrease the methane number, making the gas more prone to liquid dropout and less suitable for direct compression or liquefaction.

The key challenge with associated gas is that it usually does not meet the specifications required for sale. Parameters such as heating value, methane number, hydrocarbon dew point, and contaminant levels (e.g., CO_2, H_2S, water) often fall outside acceptable limits. As a result, it cannot simply be compressed or liquefied without prior treatment. This necessitates additional processing equipment, which increases both capital and operational costs and complicates deployment—especially in remote areas where infrastructure is minimal or non-existent.

Further complicating matters, associated gas flow rates decline over the life of the well and are influenced by factors beyond the operator's control. This makes it difficult to size treatment and monetization systems correctly and undermines the economics of traditional infrastructure-heavy solutions.

The F2X system is specifically engineered to address these challenges. It features an embedded, versatile treatment unit capable of handling rich gas variability; its mobile and scalable design allows for rapid deployment and adjustment to fluctuating flow rates. Critically, it operates as a standalone system, requiring minimal on-site infrastructure, making it ideally suited for flare gas recovery in remote and dynamic production environments.

Technology Overview

The F2X unit is a fully integrated, trailer-mounted LNG production platform designed for field mobility, modularity, and standalone operation. It consists of five core trailers, each fulfilling a critical process function:

- Gas Treatment Trailer: Performs CO_2 removal via amine absorption, dehydration via molecular sieves, and optional heavies/NGL separation depending on feed gas composition
- Compression Trailer: Boosts inlet gas to optimal pressure levels for refrigeration cycles.
- Liquefaction Trailer: Employs a refrigerant loop and brazed plate exchangers to achieve high liquefaction efficiency.
- Control & Automation Trailer: Hosts HMI panels, SCADA (supervisory control and data acquisition) architecture, and Azure IoT cloud integration for real-time monitoring, control, and diagnostics.
- Power Generation Trailer: Provides energy via a genset fueled by tail gas, ensuring off-grid capability and energy self-sufficiency.

In addition to these trailers, the F2X configuration includes:

- LNG Storage Tanks: For high-purity methane-rich LNG product
- NGL Storage Tanks: For recovered liquids, especially propane and butane fractions

These auxiliary systems are skid-mounted and placed alongside the main trailers to ensure seamless handling, storage, and distribution. The inclusion of LNG/NGL tanks enables direct-to-market energy transformation from flare gas—closing the loop from gas capture to liquid delivery.

The platform is engineered for rapid mobilization and redeployment, typically within a few weeks. Its "deploy, produce, and redeploy" design makes it highly suitable for oilfields with transient flare profiles, stranded assets, or rotational pad developments. Compared to centralized gas processing plants, F2X offers operators a fraction of the upfront investment, minimal site preparation, and faster permitting cycles. This makes it a compelling option in regions with time-sensitive or capital-constrained flare challenges. These modular designs have shown favorable economics in small-scale LNG applications across several studies (Zhou et al., 2021).

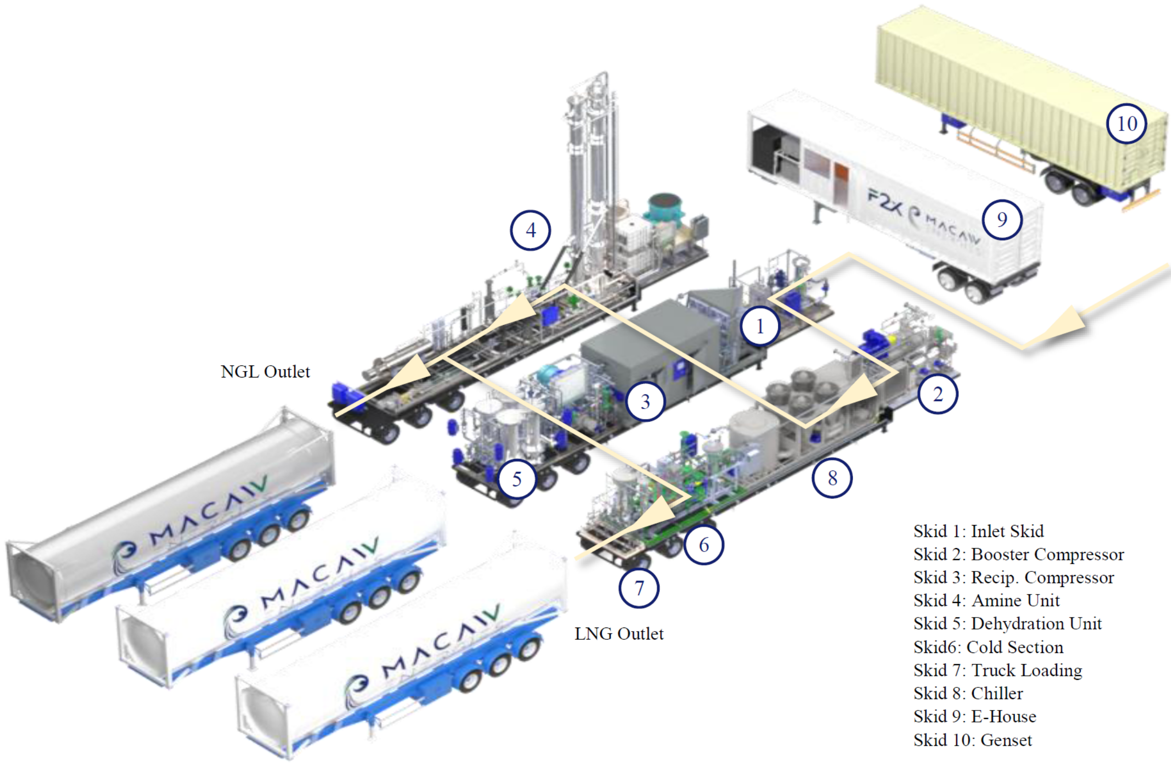


Figure. 1—MACAW F2X platform: integrated configuration for gas treatment and liquefaction.

MACAW offers two commercial models:

- F2X1000: Designed to process up to 1 MMSCFD, optimized for flare clusters or pad sites with variable volumes between 1 to 6 MMSCFD.
- F2X5000: Designed to process up to 5 MMSCFD, optimized for flare clusters or pad sites with variable volumes between 5 to 15 MMSCFD.

A digital twin is in development to support predictive maintenance, load forecasting, and CO2 accounting across fleet operations. This will enable seamless ESG reporting for operators and financiers.

Value Proposition for Oil Producers

For oil producers, F2X delivers an actionable response to flaring mandates, rising diesel costs, and ESG investment pressures. Its economic model is centered on flexibility, fast payback, and minimal operational disruption.

The F2X platform:

- Enables compliance with flaring bans and methane emission regulations
- Avoids curtailment or shut-in due to associated gas constraints
- Converts waste into revenue by creating a transportable, saleable LNG product
- Displaces expensive diesel in operations such as drilling, completions, and mobile power
- Enhances ESG profiles with a field-proven, certified low-carbon intensity solution

A single F2X1000 unit processing 1 MMSCFD can avoid over 20,000 tCO2e annually compared to diesel, while capturing value from gas that would otherwise be lost. With modular deployment and short commissioning time, F2X provides near-immediate impact without permanent infrastructure investment.

MACAW Energies offers leasing and revenue-share models to reduce CAPEX burden and accelerate ROI. For operators, this means scalable emissions reduction at minimal risk—fully aligned with both production goals and net-zero commitments.

Whether deployed for pad clusters, rotational wells, or multi-client hubs, F2X is adaptable, commercial-ready, and already operating in the field today.

Field Performance (Texas Pilot)

In 2024, an F2X1000 unit was deployed at a stranded oil well in South Texas, where no gas pipeline or grid connection was available. The site exemplified the challenges faced by many operators: high Gas-Oil Ratio (GOR) associated gas, rich in heavier hydrocarbons, and no economic path to monetization.

Figure. 2—F2X Mobilization

The F2X system was mobilized and operational within weeks, without the need for external utilities, civil works, or permanent infrastructure. By directly connecting to the wellhead, the system immediately began processing associated gas into LNG without the need for external utilities or grid access. This rapid deployment enabled the operator to maintain oil production uninterrupted and meet state-level flaring compliance.

Figure. 3—well site

During the field campaign, the unit achieved stable inlet flow between 0.5 and 1.1 MMSCFD and produced 12–15 metric tons per day of LNG. No flaring or venting occurred throughout the operation. Boil-off and tail gas were reused for power generation, enabling a closed-loop emissions profile. Startup time from cold conditions was under one hour with no unplanned shutdowns.

Environmental parameters at the site ranged from 8°C to over 45°C, confirming resilience under various ambient conditions. Noise levels remained below 65 dBA at 50 feet, meeting local standards. Product LNG was offloaded via pressure differential into ISO cryogenic tanks, ready for direct use in transportation or industrial applications.

Operating under high-heat and variable ambient conditions, the F2X unit validated full-cycle operability in real-world conditions. It treated raw associated gas with ~85% methane and ~2% CO2, removing impurities and heavy hydrocarbons before converting the gas into high-purity LNG.

No flaring or venting occurred during the operational period. All boil-off gas was recirculated or combusted internally for power, validating the system's zero-emission profile. LNG was offloaded into ISO cryogenic tanks via pressure differential, enabling direct downstream use in mobile or industrial applications.

Perhaps most critically, the operator avoided potential production curtailment due to flaring violations. In discussions with site staff, the system was described as "surprisingly plug-and-play" and "one of the few field solutions that delivered as promised."

The pilot demonstrated not only technical feasibility but also economic potential: the LNG output displaced diesel fuel that would otherwise be trucked in at high cost, reducing both operating expenses and Scope 1 emissions.

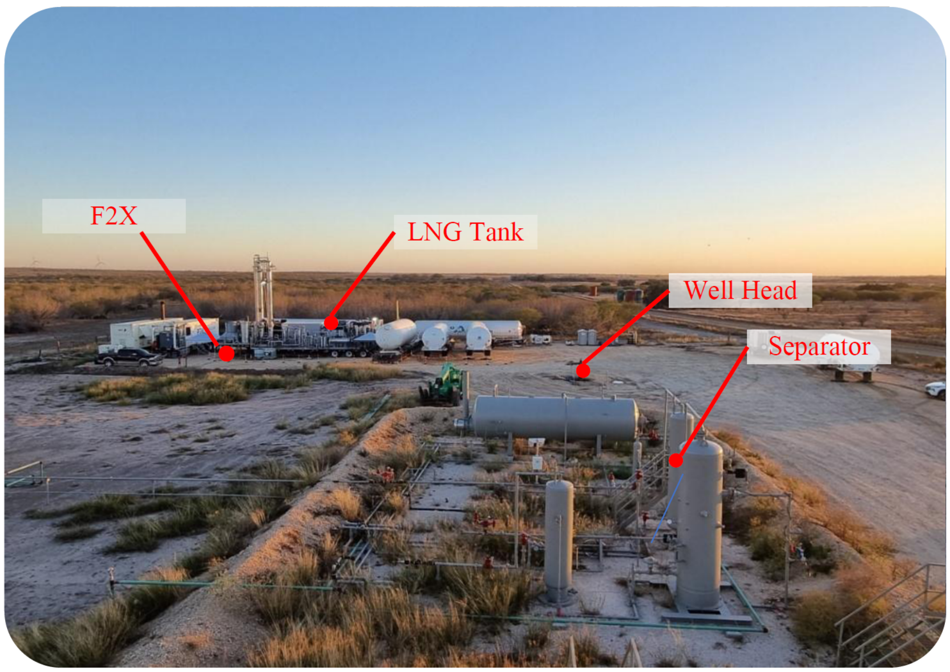


Figure. 4—F2X unit in field operation during the South Texas pilot.

This successful deployment confirmed that F2X can be rapidly deployed, remotely operated, and economically justified even at smaller wellsites. Its performance in the South Texas environment underscores its readiness for broader scale-up across North and South America, the Middle East, and other flare-prone regions.

Lifecycle Emissions and Carbon Credentials

A core differentiator of the F2X system is its independently verified climate performance. In 2024, MACAW Energies' LNG product—branded LiquidFlare—received third-party certification from SCS Global Services, using the Argonne GREET (2023) model and conforming to ISO 14067:2018 standards. The SCS certification and GREET-based modeling approach are publicly accessible through SCG Global (2024).

The lifecycle boundary spans gas acquisition, purification, liquefaction, and offloading. Emissions credits were granted based on avoided flaring, reduced methane leakage, and downstream diesel displacement. The net impact: each tonne of LNG produced results in more CO2-equivalent removed than emitted.

The verified carbon intensity of -43 kg CO_2e/MMbtu on a well-to-pump basis makes LiquidFlare one of the first negative-carbon LNG fuels validated by life cycle assessment. The assessment boundaries covered the full well-to-pump chain: gas acquisition, treatment, liquefaction, transport, and end-use substitution of diesel. This figure is based on processing ~1.1 MMSCFD of associated gas, producing ~750 MMBtu/day of LNG, and assumes transport distances of 500–800 km.

Emissions reductions were credited based on avoided flaring (CO_2 and uncombusted CH_4) and diesel displacement at the point of use. The resulting negative carbon intensity (-43 kg CO_2e/MMBtu) reflects the net environmental benefit of repurposing flare gas as a clean-burning, transportable fuel. The full LCA report and verification summary are available through SCS Global's certification registry (SCS, 2024). Similar lifecycle GHG comparisons from field studies also confirm the superior carbon performance of flare recovery over routine flaring (Shah et al., 2022).

This result reflects three main factors:

- **Elimination of flaring**: F2X prevents direct CO2 and methane emissions from combustion.
- **Use of stranded gas**: Converts waste gas into a valuable product, displacing conventional fuels.
- **Self-powered operation**: The process is powered by tail gas, minimizing external energy inputs and upstream emissions.

Each F2X1000 unit can offset approximately 20,000 metric tons of CO_2e per year. Larger configurations scale proportionally. These reductions are fully site-independent and repeatable across deployments. Emissions generated during compression and liquefaction are accounted for in the lifecycle analysis. However, they are more than offset by the avoided emissions from flaring and diesel use.

Importantly, the SCS certification opens access to voluntary carbon markets, where negative-carbon LNG may be monetized. This creates a dual incentive: commercial revenue from LNG and ESG-aligned carbon credits—making F2X attractive to both operators and financiers. As global buyers and regulators increasingly prioritize certified low-carbon fuels, this designation strengthens the commercial positioning of F2X-produced LNG.

Global Scalability

Oil and gas majors and national oil companies are publishing increasingly ambitious emission-reduction targets that directly underscore the environmental value of the F2X solution:

- OGCI (Oil & Gas Climate Initiative), which counts Saudi Aramco, BP, Chevron, CNPC, Equinor, Eni, Exxon Mobil, Oxy, Petrobras, Repsol, Shell and Total and many global operators among its members, has launched the "Aiming for Zero Methane Emissions" initiative. The collective target is to achieve near-zero methane emissions from operated assets by 2030, supported by best-practice leak detection, measurement, and transparency commitments.
- ADNOC has committed to a 25% reduction in greenhouse gas intensity by 2030 (relative to 2019), a milestone backed by plans for zero methane emissions from its operated assets by 2030, scaling carbon capture capacity to 5 Mt CO_2/yr, and advancing its goal of operational net-zero by 2045.
- Saudi Aramco participates in OGCI and its methane elimination goals and aligns with Saudi Arabia's national net-zero by 2060 target, including its pledge under the Global Methane Pledge at COP26.

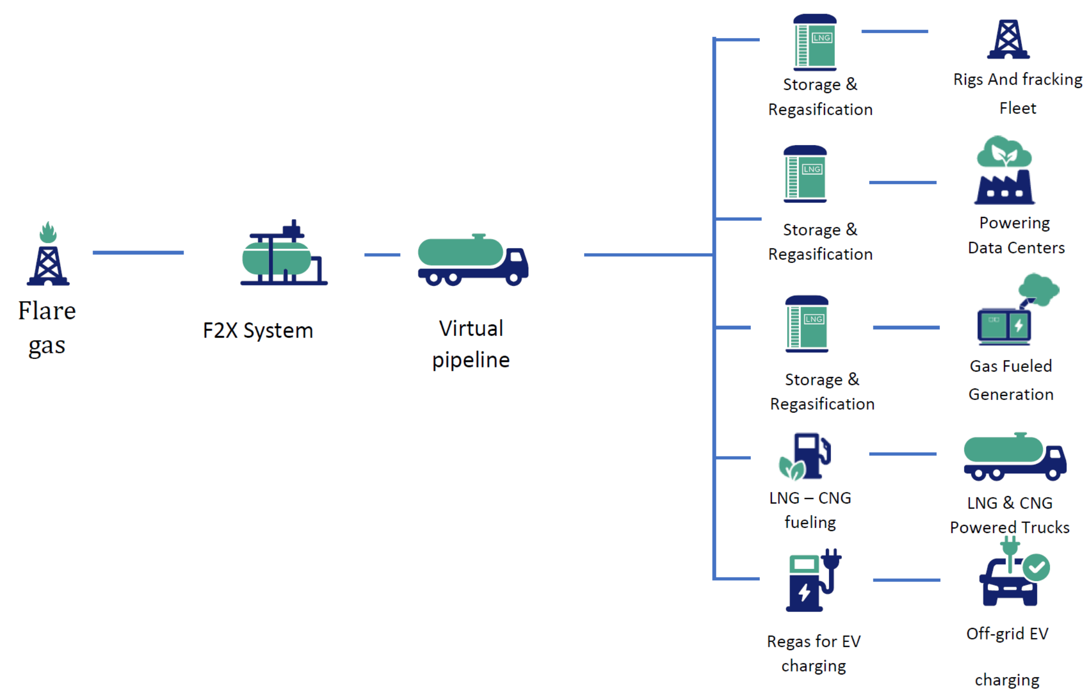
Additionally, environmental initiatives launched at COP28, such as the Oil & Gas Decarbonization Charter and the World Bank's Zero Routine Flaring by 2030, further bind the sector to eliminate routine flaring and near-eliminate methane emissions across MENA. These align closely with F2X's capacity to capture and liquefy routine flare gas, enabling operators not only to meet obligations but also to monetize compliance via verified negative carbon LNG, creating a dual benefit in both environmental performance and commercial return.

Following the pilot, MACAW has successfully validated LNG use in non-oilfield contexts, notably in powering off-grid EV fast chargers in Houston and California. In collaboration with a U.S. mobility operator, F2X delivered reliable, low-carbon LNG to replace diesel or grid-fed electricity in areas with limited infrastructure.

The LNG used in these downstream demonstrations originated from the same F2X unit deployed in South Texas, where associated gas from a stranded oil well was captured, treated, and liquefied on site. The lifecycle carbon intensity for this gas was verified by SCS Global Services at -43 kg CO_2e/MMBtu, as reported under the GREET 2023 model.

These applications underscore F2X's flexibility: it is not only a gas capture unit, but also a distributed energy enabler. Key use cases include:

- **Hydraulic fracturing fleets and drilling rigs**: LNG-powered generators displace diesel and reduce carbon intensity.
- **Construction and mining operations**: Remote operations benefit from a transportable, clean-burning fuel source.
- **Data centers & industrial power**: F2X LNG supports resilient backup power in areas with limited grid access.
- **Remote microgrids**: Combined with solar or battery storage, LNG enables hybrid off-grid systems.



One F2X1000 unit can enable up to 1,300 HP of gas engine capacity or power EV charging for up to 110 trucks per day. A 3-unit F2X5000 deployment could displace over 300,000 tons of CO_2e per year, while supporting up to 60 MW of distributed generation or fueling more than 1,500 trucks daily. Below table illustrate the energy generated from various size of F2X flare capture set-ups.

By implementing a F2X project, with a single liquefaction plant, it opens the following possibilities:

System	CO2e Reduction	Engines	Energy	LNG Truck Fueling	EV Fleet Fueling
1x F2X 1000	20,000 ton/year	Up to 1,300 HP	Up to 4 MW	Up to 40 Trucks/day	Up to 110 Trucks/day
6x F2X 1000	120,000 ton/year	Up to 7,800 HP	Up to 24 MW	Up to 240 Trucks/day	Up to 660 Trucks/day
1x F2X 5000	100,000 ton/year	Up to 6,500 HP	Up to 20 MW	Up to 200 Trucks/day	Up to 550 Trucks/day
3x F2X 5000	300,000 ton/year	Up to 19,500 HP	Up to 60 MW	Up to 600 Trucks/day	Up to 1650 Trucks/day

Globally, markets such as Vaca Muerta (Argentina), Oman, Nigeria, and the UAE present fertile ground for F2X adoption. Many of these regions face regulatory pressure to eliminate routine flaring while expanding energy access in remote regions.

In parallel, investors are increasingly rewarding field-proven emissions technologies with scalable business models. F2X offers both—verified carbon reduction and a capital-light deployment model suitable for operator-owned or third-party energy-as-a-service models.

As governments tighten methane capture mandates and buyers demand transparency in fuel lifecycle emissions, technologies like F2X will be central to future-proofing upstream operations.

Conclusion

F2X represents a pivotal advancement in how the energy sector can address one of its most persistent emissions challenges—routine flaring—while creating value through monetization of an otherwise wasted resource. It is a technology that proves emissions reduction does not have to come at the expense of operational continuity or profitability.

The F2X platform has demonstrated that field-scale micro-liquefaction is not only technically viable but commercially impactful. In its 2024 deployment in South Texas, the F2X1000 unit processed 0.5–1.1 MMSCFD of associated gas, producing 12–15 metric tons per day of high-purity LNG without any flaring or venting.

By transforming stranded gas into a flexible, low-carbon liquid fuel, F2X provides oil producers with a scalable tool to meet growing regulatory pressure, improve ESG metrics, and access emerging carbon markets. The system's mobility, modularity, and rapid deployment make it especially suited to the dynamic realities of oilfield development, offering a meaningful alternative to high-capex infrastructure or temporary flaring exemptions.

Lifecycle analysis conducted by SCS Global Services using the GREET (2023) model verified a certified carbon intensity of -43 kg CO_2e/MMBtu. This corresponds to a potential offset of over 20,000 tCO_2e per unit per year when displacing diesel, creating a clear emissions and financial case for mobile LNG deployment.

Unlike other emissions solutions that require years to deploy or deliver uncertain economics, F2X has already demonstrated real-world impact. It has been deployed, tested, certified, and commercialized in both upstream and distributed energy contexts.

As oil and gas companies seek actionable, field-ready technologies that bridge the gap between climate goals and production realities, F2X stands out as a practical, cost-efficient, and investment-aligned answer. The path forward for emissions compliance is no longer theoretical—it's mobile, it's modular, and it's already working in the field.

Acknowledgments
The authors thank the Texas Oil Company for field access and supporting the technology deployment acceleration. Additional thanks to the EV infrastructure partner for collaboration in California and Texas demonstrations.

Acronyms and Abbreviations
GOR — gas-oil ratio.
LNG — Liquefied Natural Gas
SCADA — supervisory control and data acquisition

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